

AI 2002 – Artificial Intelligence (Spring 2025)

Assignment 5

<u>Topics Covered:</u> ANN, PERCEPTRON,PERCEPTRON LEARNING RULE,MULTI LAYER PERCEPTRON , BACK PROPAGATION.	<u>Submission Deadline:</u> <i>Wednesday – May 14, 2025 by 23.00 sharp</i>
<u>Submission Guidelines:</u> • Group assignment: 2 members (at max.) or an individual • Submit your code solution for Questions 6, and a PDF file for Questions 1 to 5, on 3. the Google Classroom (of your respective section).	

Question No 01:

Consider a neural network that models the **XNOR** function using one hidden layer with two neurons and a single output neuron.

1. Use the **sigmoid** activation function and initialize all weights and biases to **0.2**. Perform the **first and second iteration of the backpropagation algorithm** using the training data below:

X1	X2	Y
0	0	1
1	0	0
0	1	0
1	1	1

2. Now, consider the same network using **ReLU** as the activation function. How will the weight updates differ in the first two iterations? Discuss and calculate.

Question No 02:

1. Derive a gradient descent based learning rule for a perceptron that uses tanh as activation function. Assume that the perceptron has D inputs and a bias term and we have N training examples (X_i, Y_i) $i = 1, 2, \dots, N$ to learn weights of the perceptron. Please note that the derivative of $\tanh(x)$ is $1 - \tanh^2(x)$

2. Show the mathematical working of Artificial Neural Network by taking the case in figure below. First two columns are the input values for X1 and X2 and the third column is the desired output. Show only 3 iterations.

Learning rate = 0.2

Threshold = 0.5

Actual output = $W_1X_1 + W_2X_2$

Next weight adjustment = $W_n + \Delta W_n$

Change in weight = $\Delta W_n = \text{learning rate} * (\text{desired output} - \text{actual output}) * X_n$

Show two complete iterations for acquiring the desired output.

X1	X2	W1	W2	D	Y	ΔW_1	ΔW_2
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X1	X2	Output
0	0	0
0	1	1
1	0	0
1	1	1

Question No 03:

You are given a feed-forward neural network with two inputs (x_1 , x_2), two hidden neurons, and one output neuron (Y). Each neuron has a bias of -1. All neurons use the tanh activation function.

Given:

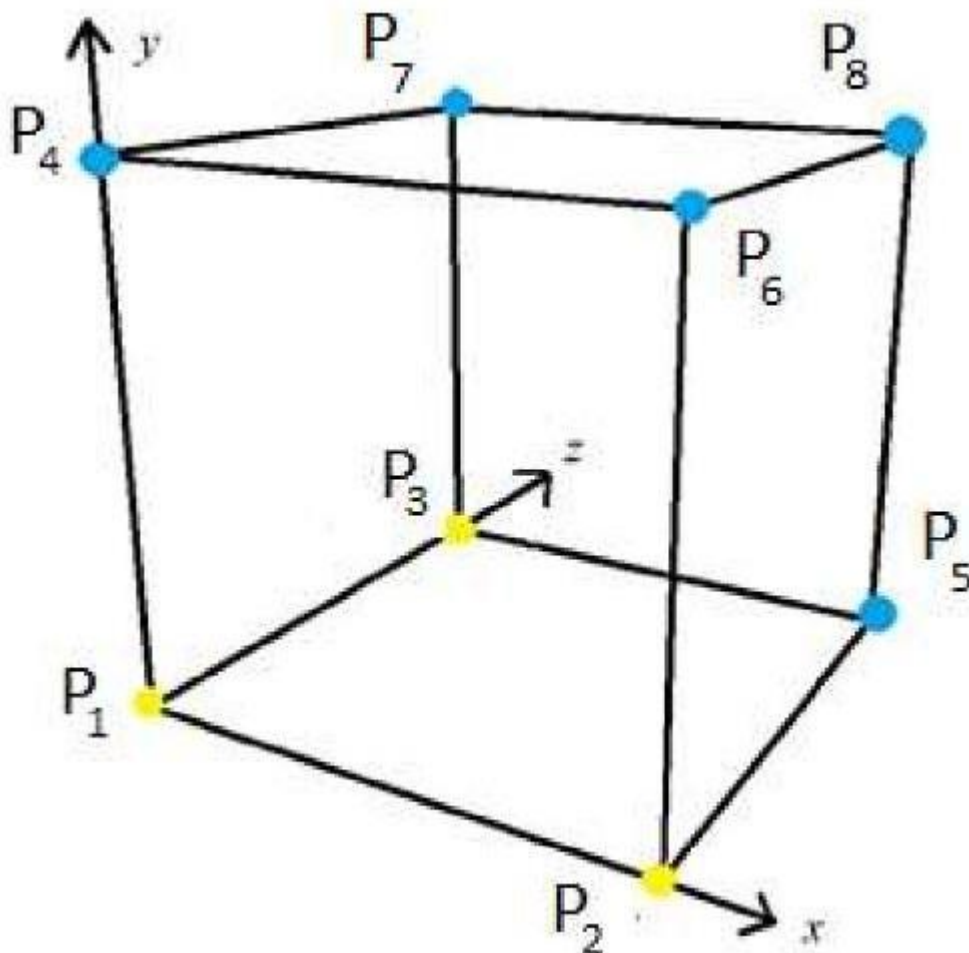
- Input: $X = [1, 0]$
- Target Output: $Y = 1$
- Initial weights as:
 - Input to Hidden: $w_{11} = 0.3$, $w_{12} = -0.2$, $w_{21} = 0.4$, $w_{22} = 0.1$
 - Hidden to Output: $v_1 = 0.5$, $v_2 = -0.4$
- Learning rate $\eta = 0.1$

Tasks:

- Perform a forward pass to compute the output.
- Compute error at the output and backpropagate it.
- Calculate δ for each neuron and update all weights using the backpropagation algorithm.

Question No 04:

You are given a 2-class, 3-dimensional data set. The data is plotted in the graph below. Yellow and blue dots represent negative class (-1) and positive class (1) respectively. Basically, the samples/points represent the corners of a unit cube. The point P1 is at the centre of the coordinate plane i.e. (0, 0, 0).



Using a single Perceptron, learn a linear model for this data that correctly classifies all the samples/points. You must start with the initial weights $[-2 \ 2 \ 3 \ 2]$ and use the points in order i.e. P1 then P2 ... P8 during the learning process. Further, the first weight i.e. -2 is that of the bias term the bias term is always 1 and learning rate is also 1. Show all the working that you did to calculate the values of the weights. The activation function $g(x)$ is given below:

$$g(x) = \begin{cases} 1, & x \geq 0 \\ -1, & x < 0 \end{cases}$$

Question No 05:

Consider a single perceptron unit with a bias term and n inputs [**x0, x1, ... xn**]. The perceptron uses weights [**w0, w1, ... wn**] and the given sigmoid activation function to compute the output.

$$\text{(i.e. } f(x) = \frac{1}{1+e^{-x}} \text{)}$$

Derive the weight update equation for such a perceptron using the following definition of

$$\textbf{Error} = \sum_j^m (y_j - y_j^*)^2 + \sum_{i=0}^n w_i^2$$

where y_j is the target output and y_j^* is the output computed using weights and the activation function of neuron.

Hints:

- Derivative of sigmoid activation function can be written as

$$f'(x) = f(x) \cdot (1 - f(x))$$

- Weights are updated using the equation

$$W_i = W_i - \eta \Delta E_i$$

Therefore, to derive the final equation you need to compute the derivative of Error with respect to each of the weights components w_i and then use the weight update equation given above to derive the final equation.

